

# Mateo Rodríguez Polo

## Doctoral Researcher in Stochastic Finance

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## Research Profile

Stochastic control, backward stochastic differential equations, time inconsistency, filtering and hidden information, mathematical finance, market microstructure, and computational methods for stochastic optimisation.

## Education

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| 2023–present | <b>PhD in Mathematics, ETH Zürich</b><br>Stochastic Finance Group. Supervisors: Prof. Dylan Possamaï and Prof. Johannes Muhle-Karbe. Research in stochastic control, mathematical finance, hidden information, and market microstructure. |
| 2021–2023    | <b>MSc in Mathematics, with distinction, ETH Zürich</b><br>GPA: 6.00/6.00. Willi Studer Prize; ETH-D Scholarship. Master's thesis supervised by Prof. Dylan Possamaï.                                                                     |
| 2020–2021    | <b>Visiting Erasmus Student, Imperial College London</b><br>Completed the final 60 ECTS of the bachelor's degree. GPA: 93.7% (top 1%).                                                                                                    |
| 2017–2021    | <b>BSc in Mathematics, Autonomous University of Madrid</b><br>GPA: 9.84/10 (top 1%). Bachelor's thesis: <i>Geometric Function Theory</i> , supervised by Prof. Dragan Vukotić; grade 10/10 with honours.                                  |
| 2006–2017    | <b>Professional Degree in Music, Professional Conservatory of Music, Salamanca</b><br>Specialisation: bassoon.                                                                                                                            |

## Professional Experience

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| Jun–Sep 2025    | <b>Quantitative Research Intern, Jump Trading, London</b><br>Ten-week quantitative-research internship.                                                                                                                        |
| Apr 2023        | <b>Quantitative Trading Intern, Jane Street Capital, London</b><br>Three-week off-cycle trading internship.                                                                                                                    |
| 2022, 2024–2025 | <b>Teaching Assistant, ETH Zürich</b><br>Problem classes in mathematical finance, applied stochastic processes, stochastics and machine learning, numerical analysis of stochastic differential equations, and linear algebra. |
| 2020–2021       | <b>Grader, Art of Problem Solving</b><br>Part-time grader specialising in advanced mathematical olympiad problems.                                                                                                             |

## Awards and Honours

- **Willi Studer Prize** for achieving a perfect score in the ETH Zürich MSc; **ETH-D Scholarship**, awarded to 30 students across all ETH master's programmes.
- **Extraordinary Award** for the highest graduating GPA at the Autonomous University of Madrid; 200 of 240 ECTS completed with honours.
- Three **Excellence Grants from the Community of Madrid**, for the academic years 2017–2018, 2018–2019, and 2019–2020.
- **Silver medal** at the 2021 International Mathematics Competition for University Students; bronze medals in 2018, 2019, and 2020.
- **Bronze medal** at the 2017 Spanish Mathematical Olympiad.
- First place at the 2017 Baccalaureate Olympiad of the University District of Salamanca; second place in 2016.
- Top-three placements in the 2012, 2013, 2014, and 2015 Secondary Education Mathematical Olympiads of Salamanca.
- Honours in the Spanish Baccalaureate; first prize in Mathematics, Physics and Chemistry at the national Young Researchers contest.

## Research Papers

- [1] D. Possamaï and **M. Rodríguez Polo**. *Here, There and Everywhere: State-Dependent Time-Inconsistent Stochastic Control*. [arXiv:2603.22022](#).
- [2] F. Beretta, D. Possamaï, **M. Rodríguez Polo**, and L. Tangpi. *Policy Iteration for Exploratory Control Problems via BSDEs*. To appear.
- [3] H. Pagès, D. Possamaï, and **M. Rodríguez Polo**. *Equilibrium Prices under Hidden Markov Fundamentals*. To appear.
- [4] J. Muhle-Karbe, R. Oomen, and **M. Rodríguez Polo**. *Information Leakage and Opportunistic Trading Around the FX Fix*. [SSRN 5351658](#).

## Conference Talks

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| Jan 2024 | <b>23rd Annual SAET Conference.</b> Santiago de Chile.                                                                          |
| Jun 2024 | <b>ETH–Hong Kong–Imperial Mathematical Finance Workshop.</b> London.                                                            |
| Jul 2024 | <b>12th World Congress of the Bachelier Finance Society.</b> Rio de Janeiro.                                                    |
| Oct 2024 | <b>Stochastic Control and Games for Risk and Regulation.</b> Hammamet.                                                          |
| Apr 2025 | <b>ETH–Hong Kong–Imperial Mathematical Finance Workshop.</b> Hong Kong.                                                         |
| Apr 2025 | <b>9th Asian Quantitative Finance Conference.</b> Shenzhen.                                                                     |
| Jun 2025 | <b>12th General AMaMeF Conference.</b> Verona.                                                                                  |
| Dec 2025 | <b>IV Workshop on Stochastic Analysis and Optimization Applied to Economics, Finance, Energy and Insurance.</b> Puerto Natales. |
| Jun 2026 | <b>Quantitative Finance Conference 2026.</b> Singapore.                                                                         |
| Jun 2026 | <b>ETH–Hong Kong–Imperial Mathematical Finance Workshop.</b> Zürich.                                                            |
| Jul 2026 | <b>13th World Congress of the Bachelier Finance Society.</b> Bologna.                                                           |

## Technical Skills and Languages

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|-------------------------|----------------------------------------------------------------|
| <b>Programming</b>      | Python, C, MATLAB, SageMath, Processing.                       |
| <b>Scientific tools</b> | L <sup>A</sup> T <sub>E</sub> X.                               |
| <b>Languages</b>        | Spanish (native); English (C2); German (B2 Goethe-Zertifikat). |

## Additional Activities

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Candidate Master in chess (title awarded in October 2022), with a FIDE rating of 2218. Competed in the Spanish First League and the Swiss National A League. Bassoonist in amateur orchestras. Interests include football, pádel, and tennis.